

**Physics 7501 Quiz 3**  
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**(2.3.1) Spin Precession**

Under a magnetic field along the  $z$ -axis, described by a Hamiltonian

$$\hat{H} = -B\hat{Z},$$

where

$$\hat{Z} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

is the 3rd Pauli matrix. Consider an initial state whose spin magnetic moment is along the  $x$ -axis. What is the spin precession periodicity for its magnetic moment? Express your answer as a function of  $B$ . You can set  $\hbar = 1$ .

Recall that the magnetic dipole moment  $\boldsymbol{\mu}$  is proportional to the spin angular momentum  $\boldsymbol{\sigma}$  (point in the same direction). Our initial state points along the  $x$ -axis (without loss of generality assume  $+x$ -direction) thus

$$|\psi(0)\rangle = |+x\rangle = \frac{1}{\sqrt{2}} (|+z\rangle + |-z\rangle)$$

$\Rightarrow$

$$|\psi(t)\rangle = \frac{1}{\sqrt{2}} \left( e^{iBt} | +z \rangle + e^{-iBt} | -z \rangle \right)$$

We want to find  $T$  such that  $|\psi(T)\rangle = |\psi(0)\rangle$ , thus

$$\frac{1}{\sqrt{2}} \left( e^{iBT} | +z \rangle + e^{-iBT} | -z \rangle \right) = \frac{1}{\sqrt{2}} (| +z \rangle + | -z \rangle)$$

$\Rightarrow$

$$\frac{1}{\sqrt{2}} \left( | +z \rangle + e^{-2iBT} | -z \rangle \right) = \frac{1}{\sqrt{2}} (| +z \rangle + | -z \rangle)$$

$\Rightarrow e^{-2iBT} = 1 \Rightarrow 2BT = 2\pi \Rightarrow T = \frac{\pi}{B}$ . Since time should be positive, we take the absolute value

$$T = \frac{\pi}{|B|}$$

**(2.3.2) Neutrino oscillation**

Both neutrino oscillation and spin precession are described by 2-level Hamiltonians of the same form. Consider two “mass eigenstates”  $|\nu_1\rangle$  and  $|\nu_2\rangle$  in neutrino oscillation: do they correspond to

Pauli- $\hat{Z}$  eigenstates or Pauli- $\hat{X}$  eigenstates in the spin precession problem? Explain.

$|\nu_1\rangle$  and  $|\nu_2\rangle$  are analogous to the Pauli- $\hat{Z}$  eigenstates since they are energy eigenstates of their associated Hamiltonian.

### (2.3.3) 2-qubit Hamiltonian

Find the smallest eigenvalue of the 2-qubit Hamiltonian

$$\hat{H} = \hat{Z}_1 \otimes \hat{Z}_2.$$

For any two operators  $\hat{A}$  and  $\hat{B}$ , if  $\lambda_a$  is an eigenvalue of  $\hat{A}$  and  $\lambda_b$  is an eigenvalue of  $\hat{B}$  i.e.  $\hat{A}|a\rangle = \lambda_a|a\rangle$  and  $\hat{B}|b\rangle = \lambda_b|b\rangle$ , then

$$(\hat{A} \otimes \hat{B})(|a\rangle \otimes |b\rangle) = \hat{A}|a\rangle \otimes \hat{B}|b\rangle = \lambda_a \lambda_b (|a\rangle \otimes |b\rangle)$$

$\implies \lambda_a \lambda_b$  is an eigenvalue of  $\hat{A} \otimes \hat{B}$  (pairwise products of the eigenvalues of  $\hat{A}$  and  $\hat{B}$ ). Since the eigenvalues of  $\hat{Z}$  are  $\pm 1$ , the eigenvalues of  $\hat{H}$  are also  $\pm 1 \implies \lambda_{\min} = -1$